PROBABILITY III

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Contents

Chapter 1

REFINING THE NOTION OF PROBABILITY

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Consider the random walk on the line; a drunk ant sits at the origin and takes steps of length 1 to the left or right with equal probability. We wish to obtain the probability that after an even n steps, the ant returns to the origin. Of course, the natural approach is to frame this problem in a discrete probability space.

Definition 1.1. A *discrete probability space* is a pair (Ω, q) where Ω is a non-empty finite or countably infinite set called the sample space, and $q : \Omega \to [0,1]$ is a function satisfying $\sum_{\omega\in\Omega} q(\omega) = 1$. Any subset of Ω is termed an event, and the probability of an event $A \subseteq \Omega$ is defined as $\mathbb{P}(A) = \sum_{\omega\in A} q(\omega)$.

For the problem above, we can take the sample space to be

$$\Omega = \{\mathbf{x} = (x_0 = 0, x_1, x_2, \dots, x_n) \in \mathbb{Z}^{n+1} \mid x_{i+1} - x_i \in \{-1, 1\} \text{ for all } 0 \le i < n\}, \tag{1.1}$$

and the mapping q to be $q(\mathbf{x}) = 2^{-n}$ for all $\mathbf{x} \in \Omega$. Then, the event that the ant returns to the origin after n steps is given by $A = \{\mathbf{x} \in \Omega \mid x_n = 0\}$, and we have

$$\mathbb{P}(A) = \sum_{\mathbf{x}\in A} q(\mathbf{x}) = 2^{-n} |A| = 2^{-n} \binom{n}{n/2}. \tag{1.2}$$

Reframing the problem, let us consider a random walk on the line starting at the origin, and continuing indefinitely. In this case, the sample space is given by

$$\Omega = \{\mathbf{x} = (x_0 = 0, x_1, x_2, \dots) \in \mathbb{Z}^{\mathbb{N}} \mid x_{i+1} - x_i \in \{-1, 1\} \text{ for all } i \in \mathbb{N}\}. \tag{1.3}$$

This Ω is uncountably infinite. In such a case, we cannot define a probability measure by assigning a positive probability to each element of Ω. As another example of difficulty in dealing with uncountable sample spaces, consider the problem of choosing a real number uniformly at random from the interval $[0,1]$. If we attempt to assign a positive probability to each element of $[0,1]$, we would have to assign a probability of 0 to each element, since there are uncountably many elements in $[0,1]$. However, this would lead to a total probability of 0, which is not valid. Of course, we know that the probability of choosing a specific element or a countable set of elements from $[0,1]$ is 0, but there then arises the question of how to assign a probability to an uncountable set of elements, such as the Cantor set. We can work in a slightly different viewpoint.

Definition 1.2. A probability space is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is a non-empty set called the sample space, $\mathcal{F}$ is the collection of all subsets of Ω $(\sim 2^{\Omega})$, and $\mathbb{P} : \mathcal{F} \to [0,1]$ is a function satisfying $\mathbb{P}(\Omega) = 1$ and countable additivity, that is, for any countable collection of pairwise disjoint sets $A_1, A_2, \dots \in \mathcal{F}$, we have

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i). \tag{1.4}$$

Using this definition, we look to define a $\mathbb{P}$ on subsets of $[0,1]$ that satisfies the above properties, along with the property that $\mathbb{P}([a,b]) = b-a$. Thus, we have $\mathbb{P}([0,1]) = 1$. If we have two disjoint intervals $I_1, I_2 \subseteq [0,1]$, then $\mathbb{P}(I_1 \cup I_2) = \mathbb{P}(I_1) + \mathbb{P}(I_2)$. This is a natural property to expect from a probability measure. What about a general set $S \subseteq [0,1]$? One can observe that if we have a sequence of intervals $I_1, I_2, \ldots$ such that $S \subseteq \bigcup_{k=1}^{\infty} I_k$, then we would have $\mathbb{P}(S) \leq \sum_{k=1}^{\infty} \mathbb{P}(I_k)$. This is because the probability of S should be less than or equal to the probability of any set that contains S. Thus, we can define

$$\mathbb{P}_*(S) = \inf\left\{\sum_{k=1}^{\infty} |I_k| \;\middle|\; S \subseteq \bigcup_{k=1}^{\infty} I_k,\ I_k \subseteq [0,1]\right\}. \tag{1.5}$$

We have not verified that this is a valid definition of a probability measure, hence the subscript $*$. Via this, let us try computing $\mathbb{P}_*(\mathbb{Q} \cap [0,1])$. Enumerate the rationals in $[0,1]$ as $\{q_1, q_2, \ldots\}$. For any $\epsilon > 0$, we can choose intervals $I_k = [q_k - \epsilon/2^{k+1}, q_k + \epsilon/2^{k+1}] \cap [0,1]$ for each $k \in \mathbb{N}$. Note that $\bigcup_{k=1}^{\infty} I_k$ contains all the rationals in $[0,1]$, and $\sum_{k=1} |I_k| \leq \sum_{k=1}^{\infty} \epsilon/2^k = \epsilon$. Since ϵ was arbitrary, we have $\mathbb{P}_*(\mathbb{Q} \cap [0,1]) = 0$. This is consistent with our intuition that the probability of choosing a rational number from $[0,1]$ is 0. So, $\mathbb{P}_*$ seems to be a reasonable definition of a probability measure on $([0,1], 2^{[0,1]})$. However, we will show later that there exists a set $\mathcal{A} \subseteq [0,1]$ such that $\mathbb{P}_*(\mathcal{A}) = \mathbb{P}_*([0,1] \setminus \mathcal{A}) = 1$, contradicting the property that $\mathbb{P}_*([0,1]) = 1$. We state the following theorem without proof.

Theorem 1.3. *There is no probability measure $\mathbb{P}$ on $([0,1], 2^{[0,1]})$ such that $\mathbb{P}([a,b]) = b-a$ for all $[a,b] \subseteq [0,1]$.*

If we remove the requirement that the probability measure of a subinterval $[a,b]$ is equal to its length, then we can define a probability measure on $([0,1], 2^{[0,1]})$. For example, we can define $\mathbb{P}(S) = 1$ if $1/2 \in S$ and $\mathbb{P}(S) = 0$ otherwise. This is a valid probability measure, but it is not very useful. Keeping the requirement that the probability measure of a subinterval $[a,b]$ is equal to its length, we can instead restrict the collection of subsets of $[0,1]$ on which we define the probability measure.

Definition 1.4. A *probability space* is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$ where

- Ω is a non-empty set called the sample space,
- $\mathcal{F}$ is a collection of subsets of Ω satisfying the following properties:
  - $\Omega \in \mathcal{F}$,
  - if $A \in \mathcal{F}$, then $\Omega \setminus A \in \mathcal{F}$,
  - if $A_1, A_2, \ldots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$,
- $\mathbb{P} : \mathcal{F} \to [0,1]$ is a function satisfying:
  - $\mathbb{P}(\Omega) = 1$,
  - if $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint, then $\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

Such an $\mathcal{F}$ is termd a *σ-algebra* on Ω, and such a $\mathbb{P}$ is termed a *probability measure* on $(\Omega, \mathcal{F})$.

1.1 σ-algebras

Example 1.5. Given Ω, trivial possible choices for $\mathcal{F}$ are $\{\emptyset, \Omega\}$ and 2^{Ω}. A more interesting example is the following: suppose Ω is uncountable. Then $\mathcal{F} = \{A \subseteq \Omega \mid A \text{ is countable or } \Omega \setminus A \text{ is countable}\}$ is a σ-algebra on Ω.

Let $S \subseteq 2^{\Omega}$ be a collection of subsets of Ω. We claim that there exists a notion of the smallest σ-algebra containing S. This is called the σ-algebra generated by S, and is denoted by $\sigma(S)$.

Lemma 1.6. *For any collection $S \subseteq 2^{\Omega}$, there exists a unique smallest σ-algebra containing S satisfying*

$$\sigma(S) = \bigcap \{\mathcal{F} \subseteq 2^{\Omega} \mid \mathcal{F} \text{ is a } \sigma\text{-algebra and } S \subseteq \mathcal{F}\}. \tag{1.6}$$

Proof. It is easy to check that $\sigma(S)$ is indeed a σ-algebra containing S. If $\mathcal{F}$ is any σ-algebra containing S, then $\sigma(S) \subseteq \mathcal{F}$. Thus, $\sigma(S)$ is the unique smallest σ-algebra containing S. ■

Definition 1.7. Let (X, d) be a metric space (or a topological space), and let $\mathcal{S} \subseteq 2^X$ be the colletion of all open sets in X. Then, the σ-algebra $\sigma(\mathcal{S})$ is called the *Borel σ-algebra* on X, and its elements are called *Borel sets*.

We will show that there exists a probability measure on $([0,1], \mathcal{B})$, where $\mathcal{B}$ is the Borel σ-algebra on $[0,1]$, which extends the notion of length of intervals to Borel sets.

Let us return to the problem of indefinite random walks on the line. Let $\Omega = \{\omega = (\omega_1, \omega_2, \ldots) \mid \omega_i \in \{0,1\}\}$, where $\omega_i = 0$ corresponds to a step to the left and $\omega_i = 1$ corresponds to a step to the right. If we let $A = \{0,1\}$, then $\Omega = A^{\mathbb{N}}$; consider the discrete topology on A, and the product topology on Ω. If we let

$$\mathcal{S} = \{U_1 \times U_2 \times \cdots \times U_k \times A \times A \cdots \mid U_i = \{0\} \text{ or } \{1\}\} \tag{1.7}$$

then one can show that the σ-algebra generated by $\mathcal{S}$ is the Borel σ-algebra on Ω. We can set $\mathcal{F} = \sigma(\mathcal{S})$, and define a probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ by setting $\mathbb{P}(U_1 \times U_2 \times \cdots \times U_k \times A \times A \cdots) = 2^{-k}$ for all $U_i = \{0\}$ or $\{1\}$.

July 24th.

Example 1.8. Let (X, d) be a separable metric space (that is, there exists a countable dense subset). The Borel σ-algebra on X is then

$$\mathcal{B}(X) = \sigma(\{B(p,r)\}_{p \in X, r > 0}). \tag{1.8}$$

Note that the usual containment is $\sigma(\{B(p,r)\}_{p \in X, r>0}) \subseteq \mathcal{B}(X)$ for a metric space X. So, it suffices to show that if $V \in \tau$, the topology on X, then $V \in \sigma(\{B(p,r)\}_{p \in X, r>0})$. Let Q be a countable dense subset of X. Then, for any $V \in \tau$, we have

$$V = \bigcup_{n=1}^{\infty} B(q_n, r_n), \tag{1.9}$$

where $q_n \in Q \cap V$ and $r_n > 0$ (where r_n is the largest possible radius such that $B(q_n, r_n) \subseteq V$). Since $\sigma(\{B(p,r)\}_{p \in X, r>0})$ is a σ-algebra, we have $V \in \sigma(\{B(p,r)\}_{p \in X, r>0})$. Thus, we have $\mathcal{B}(X) = \sigma(\{B(p,r)\}_{p \in X, r>0})$.

Example 1.9. Let $(\Sigma, \mathcal{F}, \mathbb{P})$ be a probability space.

- $\mathcal{F}$ is closed under countable unions, countable intersections, differences, and symmetric differences. Countable unions and intersections follow from the definition of a σ-algebra. If $A, B \in \mathcal{F}$, then $A \setminus B = A \cap (\Omega \setminus B) \in \mathcal{F}$. Finally, $A \triangle B = (A \setminus B) \cup (B \setminus A) \in \mathcal{F}$.
- Let $(\Sigma, \mathcal{F}, \mathbb{P})$ be a probability space, and let $A_1, A_2, \ldots \in \mathcal{F}$. Define

$$\limsup_{n \to \infty} A_n := \{x \in \Omega \mid x \text{ belongs to infinitely many } A_n\} \tag{1.10}$$

and

$$\liminf_{n \to \infty} A_n := \{x \in \Omega \mid x \text{ belongs to all but finitely many } A_n\}. \tag{1.11}$$

Then, both the lim sup and lim inf belong in $\mathcal{F}$. To see this, simply note that

$$\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} A_k, \quad \liminf_{n\to\infty} A_n = \bigcup_{n=1}^{\infty}\bigcap_{k=n}^{\infty} A_k. \tag{1.12}$$

- One may show that $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(A) \leq \mathbb{P}(B)$ if $A \subseteq B$, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, and $\mathbb{P}(\bigcup A_n) \leq \sum \mathbb{P}(A_n)$ for any countable collection of sets $A_1, A_2, \ldots \in \mathcal{F}$.
- For a nested sequence of sets $A_1 \subseteq A_2 \subseteq \cdots$ in $\mathcal{F}$, let $A = \bigcup_{n=1}^{\infty} A_n$ (this is usually denoted by $A_n \uparrow A$). Then, $\mathbb{P}(A) = \lim_{n\to\infty}\mathbb{P}(A_n)$. To this end, let $B_n = A_n \setminus A_{n-1}$ for $n > 1$, and $B_1 = A_1$. Then $A = \bigcup_{n=1}^{\infty} B_n$. Thus,

$$\mathbb{P}(A) = \sum_{n=1}^{\infty}\mathbb{P}(B_n) = \lim_{N\to\infty}\sum_{n=1}^{N}\mathbb{P}(B_n) = \lim_{N\to\infty}\sum_{n=2}^{N}\mathbb{P}(A_n) - \mathbb{P}(A_{n-1}) + \mathbb{P}(A_1) = \lim_{N\to\infty}\mathbb{P}(A_N). \tag{1.13}$$

1.1.1 Other Systems of Sets

The following are other interesting collection of subsets of Ω besides the Borel σ-algebra.

1. *π-system*: a collection of subsets $S \subseteq \Omega$ such that $A, B \in S$ implies $A \cap B \in S$.
2. *λ-system*: a collection Λ of subsets of Ω such that $\Omega \in \Lambda$, if $A, B \in \Lambda$ with $A \subseteq B$, then $B \setminus A \in \Lambda$, and if $A_1, A_2, \ldots \in \Lambda$, then $\bigcup_{n=1}^{\infty} A_n \in \Lambda$.
3. *algebra*: a collection $\mathcal{A}$ of subsets of Ω such that $\Omega \in \mathcal{A}$, if $A \in \mathcal{A}$, then $\Omega \setminus A \in \mathcal{A}$, and if $A, B \in \mathcal{A}$, then $A \cup B \in \mathcal{A}$.

We had defined $\omega(S)$, the σ-algebra generated by a collection of subsets $S \subseteq 2^{\Omega}$. We can similarly define $\pi(S)$, the π-system generated by S, $\lambda(S)$, the λ-system generated by S, and $\mathcal{A}(S)$, the algebra generated by S.

Theorem 1.10 (*Sierpiński-Dynkin π-λ Theorem*)**.** *Let $S, \mathcal{F} \subseteq 2^{\Omega}$ be collections of subsets of Ω. Then,*

1. *$\mathcal{F}$ is a σ-algebra if and only if $\mathcal{F}$ is both a π-system and a λ-system.*
2. *If S is a π-system, then $\lambda(S) = \sigma(S)$.*

Proof.

1. If $\mathcal{F}$ is a σ-algebra, then it is trivially both a π-system and a λ-system. Conversely, if $\mathcal{F}$ is both a π-system and a λ-system, we need to verify the axioms of a σ-algebra. We have $\Omega \in \mathcal{F}$ since $\mathcal{F}$ is a λ-system. If $A \in \mathcal{F}$, then $\Omega \setminus A \in \mathcal{F}$ since $\mathcal{F}$ is a λ-system. Finally, if $A_1, A_2, \ldots \in \mathcal{F}$, then we can write $B_n = \bigcup_{i=1}^{n} A_i \in \mathcal{F}$ since $A_i^c \in \mathcal{F}$ and $B_n^c = \bigcap_{i=1}^{n} A_i^c \in \mathcal{F}$ since $\mathcal{F}$ is a π-system. Then, $B_n \uparrow B = \bigcup_{i=1}^{\infty} A_i$, and since $\mathcal{F}$ is a λ-system, we have $B \in \mathcal{F}$. Thus, $\mathcal{F}$ is a σ-algebra.
2. Let S be a π-system. Since $\lambda(S) \subseteq \sigma(S)$, it suffices to show that $\lambda(S)$ is a π-system. Fix $A \in S$, and define $\mathcal{C}_A = \{B \in \lambda(S) \mid A \cap B \in \lambda(S)\}$. Note that $S \subseteq \mathcal{C}_A$ because S is a π-system. We show that $\mathcal{C}_A$ is a λ-system. We have $\Omega \in \mathcal{C}_A$ since $A \cap \Omega = A \in \lambda(S)$. If $B_1, B_2 \in \mathcal{C}_A$ with $B_1 \subseteq B_2$, then $A \cap B_1 \subseteq A \cap B_2$, and since $A \cap B_2 \in \lambda(S)$, we have $(A \cap B_2) \setminus (A \cap B_1) = A \cap (B_2 \setminus B_1) \in \lambda(S)$, and thus $B_2 \setminus B_1 \in \mathcal{C}_A$. Finally, if $B_1, B_2, \ldots$ are in $\mathcal{C}_A$, then $A \cap B_n \in \lambda(S)$ for all n, and thus $\bigcup(A \cap B_n) = A \cap (\bigcup B_n) \in \lambda(S)$, and hence $\bigcup B_n \in \mathcal{C}_A$. Thus, $\mathcal{C}_A$ is a λ-system. Since $\lambda(S)$ is the smallest λ-system containing S, we have $\lambda(S) = \mathcal{C}_A$, and hence for any $B, C \in S$, we have $B, C \in S$ implies $B \cap C = A \cap C \in S$. Thus, $\lambda(S)$ is a π-system.

■

Lemma 1.11. *Let S be a π-system, and let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures on $(\Sigma, \sigma(S))$ such that $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in S$. Then, $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in \sigma(S)$.*

Proof. Let $\mathcal{C} = \{B \in \sigma(S) \mid \mathbb{P}(B) = \mathbb{Q}(B)\}$. Then, $S \subseteq \mathcal{C}$, and we show that $\mathcal{C}$ is a λ-system. We have $\Omega \in \mathcal{C}$ since $\mathbb{P}(\Omega) = 1 = \mathbb{Q}(\Omega)$. If $B_1, B_2 \in \mathcal{C}$ with $B_1 \subseteq B_2$, then $\mathbb{P}(B_2 \setminus B_1) = \mathbb{P}(B_2) - \mathbb{P}(B_1) = \mathbb{Q}(B_2) - \mathbb{Q}(B_1) = \mathbb{Q}(B_2 \setminus B_1)$, and hence $B_2 \setminus B_1 \in \mathcal{C}$. Finally, if $B_1, B_2, \ldots$ are in $\mathcal{C}$, then $\mathbb{P}(A) = \lim_{n\to\infty} \mathbb{P}(A_n)$, and since $\mathbb{Q}(A) = \lim_{n\to\infty} \mathbb{Q}(A_n)$, we have $\mathbb{P}(A) = \mathbb{Q}(A)$. ■

July 28th.

Example 1.12. We wish to obtain an example of S which is *not* a π-system, and two probability measures $\mathbb{P}$ and $\mathbb{Q}$ on $(\Sigma, \sigma(S))$ such that $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in S$, but $\mathbb{P}(A) \neq \mathbb{Q}(A)$ for some $A \in \sigma(S)$. Let $\Omega = \{1, 2, 3, 4\}$, and let $S = \{\{1, 2\}, \{2, 3\}\}$. Then, $\sigma(S) = 2^\Omega$. Define $\mathbb{P}$ and $\mathbb{Q}$ on $(\Omega, \sigma(S))$ by

$$\mathbb{P}(\{1\}) = \mathbb{P}(\{2\}) = \mathbb{P}(\{3\}) = \mathbb{P}(\{4\}) = 1/4, \tag{1.14}$$

and

$$\mathbb{Q}(\{1\}) = \mathbb{Q}(\{3\}) = 1/6, \quad \mathbb{Q}(\{2\}) = \mathbb{Q}(\{4\}) = 1/3. \tag{1.15}$$

Then, $\mathbb{P}(\{1, 2\}) = \mathbb{Q}(\{1, 2\}) = 1/2$ and $\mathbb{P}(\{2, 3\}) = \mathbb{Q}(\{2, 3\}) = 1/2$, but $\mathbb{P}(\{3\}) = 1/4 \neq 1/6 = \mathbb{Q}(\{3\})$.

Example 1.13. Let $\Omega \neq \emptyset$, and A_λ be a collection of subsets of Ω indexed by $\lambda \in \Lambda$. Consider the σ-algebra $\sigma(\{A_\lambda\}_{\lambda\in\Lambda})$, and pick $B \in \sigma(\{A_\lambda\}_{\lambda\in\Lambda})$. We will show that there exists a countable subcollection $\{A_{\lambda_n}\}_{n=1}^\infty$ such that $B \in \sigma(\{A_{\lambda_n}\}_{n=1}^\infty)$. Let $\mathcal{C} = \{B \in \sigma(\{A_\lambda\}_{\lambda\in\Lambda}) \mid B \in \sigma(\{A_{\lambda_n}\}_{n=1}^\infty)$ for some countable subcollection $\{A_{\lambda_n}\}_{n=1}^\infty\}$. Then, $\mathcal{C}$ is a σ-algebra containing $\{A_\lambda\}_{\lambda\in\Lambda}$, and hence $\sigma(\{A_\lambda\}_{\lambda\in\Lambda}) \subseteq \mathcal{C}$. Thus, $B \in \mathcal{C}$, and hence there exists a countable subcollection $\{A_{\lambda_n}\}_{n=1}^\infty$ such that $B \in \sigma(\{A_{\lambda_n}\}_{n=1}^\infty)$.

1.2 The Lebesgue Measure

Theorem 1.14. *There exists a unique Borel measure λ on $[0, 1]$ such that $\lambda(I) = |I|$ for all intervals $I \subseteq [0, 1]$.*

Proof. For the π-system $S = \{(a, b] \mid 0 \leq a < b \leq 1\}$, we know that $\sigma(S)$ is a Boreal σ-algebra on $[0, 1]$. If λ and μ are two Borel measures on $[0, 1]$ such that $\lambda(I) = \mu(I) = |I|$ for all intervals $I \subseteq [0, 1]$, then by the previous lemma, we have $\lambda(A) = \mu(A)$ for all $A \in \sigma(S)$. Thus, the Borel measure is unique.

To show the existence of such a measure, we will define a pseudomeasure λ_* that satisfies the interval property, and then show that it is a measure on the Borel σ-algebra. For any $E \subseteq [0, 1]$, define

$$\lambda_*(E) = \inf\left\{\sum_{i=1}^\infty |I_i| \;\middle|\; I_i \text{ are intervals and } E \subseteq \bigcup_{i=1}^\infty I_i\right\} \tag{1.16}$$

Infer that $0 \leq \lambda_*(E) \leq 1$ for all $E \subseteq [0, 1]$. Moreover, let A and B be two sets in $[0, 1]$. Then, for any $\varepsilon > 0$, we have

$$\lambda_*(A \cup B) \leq \sum_{k=1}^\infty |I_k| + \sum_{k=1}^\infty |J_k| \leq \lambda_*(A) + \varepsilon + \lambda_*(B) + \varepsilon \implies \lambda_*(A \cup B) \leq \lambda_*(A) + \lambda_*(B). \tag{1.17}$$

Finally, let $(a, b] \subseteq [0, 1]$ be an interval. Then, $\lambda_*((a, b]) = b - a$. To see this, suppose $(a, b] \subseteq \bigcup_{k=1}^\infty I_k$ for some intervals I_k. Now consider $[a + \varepsilon, b]$ for some $\varepsilon > 0$ such that $[a + \varepsilon, b] \subseteq \bigcup_{k=1}^\infty I_k$. This is a compact subinterval, so we can extract a finite subcover $[a + \varepsilon, b] \subseteq \bigcup_{k=1}^N I_k$. Then,

$$b - a - \varepsilon = |[a + \varepsilon, b]| \leq \sum_{k=1}^N |I_k| \leq \sum_{k=1}^\infty |I_k| \implies b - a \leq \lambda_*((a, b]). \tag{1.18}$$

Now, since $(a, b] \subseteq (a, b+\varepsilon)$, a similar line of reasoning shows that $\lambda_*((a, b]) \leq b-a$. Thus, we have $\lambda_*((a, b]) = b - a$. The property of finite subadditivity also implies the property of countable subadditivity (show this). We now construct the σ-algebra $\mathcal{F}$ on which λ_* is a measure. We pick

$$\mathcal{F} = \{A \subseteq [0,1] \mid \lambda_*(E) = \lambda_*(E \cap A) + \lambda_*(E \cap A^c) \text{ for all } E \subseteq [0,1]\}. \tag{1.19}$$

We show that $\mathcal{F}$ is a σ-algebra containing the Borel σ-algebra, and that λ_* is a measure on $\mathcal{F}$. It is left as a minor exercise to show that $\mathcal{F}$ is closed under intersections and complements (making $\mathcal{F}$ a π-system). We show $\mathcal{F}$ is a λ-system. Let $A, B \in \mathcal{F}$. Then $B \setminus A = B \cap A^c$ is in $\mathcal{F}$. Then, for any $E \subseteq [0,1]$, we have

$$\lambda_*(E) = \lambda_*(E \cap A_n) + \lambda_*(E \cap A_n^c) \tag{1.20}$$
$$\geq \lambda_*(E \cap A_n) + \lambda_*(E \cap A^c), \tag{1.21}$$

because $A^c \subseteq A_n^c$. Since $A_n \uparrow A$, we have $E \cap A_n \uparrow E \cap A$, and hence

$$\lambda_*(E \cap A_n) \uparrow \lambda_*(E \cap A).$$

Passing to the limit gives

$$\lambda_*(E) \geq \lambda_*(E \cap A) + \lambda_*(E \cap A^c).$$

On the other hand, subadditivity of the outer measure gives

$$\lambda_*(E) \leq \lambda_*(E \cap A) + \lambda_*(E \cap A^c).$$

Therefore,

$$\lambda_*(E) = \lambda_*(E \cap A) + \lambda_*(E \cap A^c),$$

so $A \in \mathcal{F}$. ■

July 30th.

Note that $\mathcal{F} \subsetneq 2^{[0,1]}$; that is, there exist subsets of $[0,1]$ that are not in $\mathcal{F}$. We shall assume that $\lambda_*(E) = \lambda_*(E + x)$ for all $E \subseteq [0,1]$ and $x \in \mathbb{R}$ such that $E + x \subseteq [0,1]$. This is a reasonable assumption, since we expect the Lebesgue measure to be translation invariant (one can prove this). Moreover, instead of looking at $[0,1]$ we shall look at $([0,1), \oplus)$, where $\oplus$ is addition modulo 1. Now suppose we manage to break $[0,1)$ into countably many disjoint pieces $A_1, A_2, \ldots$ such that $\bigcup_{n=1}^{\infty} A_n = [0,1)$, and $A_i = A_1 \oplus r_i$ for some $r_i \in [0,1)$. Then, we would have $\lambda_*(A_i) = \lambda_*(A_1)$ for all i, and hence $\lambda_*([0,1)) = \sum_{n=1}^{\infty} \lambda_*(A_n) = \sum_{n=1}^{\infty} \lambda_*(A_1)$. This is a contradiction, since $\lambda_*([0,1)) = 1$ and $\sum_{n=1}^{\infty} \lambda_*(A_1)$ is either 0 or ∞. Thus, we have shown that there exists a set $A \subseteq [0,1)$ such that $A \notin \mathcal{F}$.

To construct these A_i's, simply look at the equivalence relation $\sim$ on $[0,1)$ defined by $x \sim y$ if and only if $x - y \in \mathbb{Q}$. Then, we can pick a representative from each equivalence class to form a set A_1. Then, we can define $A_i = A_1 \oplus r_i$ for some $r_i \in [0,1)$ such that $\{r_i\}_{i=1}^{\infty}$ is a complete set of representatives of $\mathbb{Q} \cap [0,1)$. This gives us the desired decomposition of $[0,1)$ into disjoint pieces.

1.2.1 Carathéodory's Extension Theorem

August 4th.

Suppose Ω is non-empty, and $\mathcal{A}$ is an *algebra*; that is, $\mathcal{A}$ is a collection of subsets of Ω such that $\Omega \in \mathcal{A}$, and $\mathcal{A}$ is closed under complements and finite unions. Further say that μ is a *pre-measure* on $\mathcal{A}$; that is, $\mu : \mathcal{A} \to [0,1]$ satisfies $\mu(\emptyset) = 0$, and if $A_1, A_2, \ldots \in \mathcal{A}$ are pairwise disjoint, then $\mu(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$. Then, we can extend μ to a measure on the σ-algebra generated by $\mathcal{A}$. This is the content of *Carathéodory's extension theorem*.

Remark 1.15.
1. Note that since $\mathcal{A}$ is a π-system, such an extension must be unique.
2. For a π-system S, a pre-measure on § gives a pre-measure on $\mathcal{A}(S)$, the algebra generated by S, since every set $B \in \mathcal{A}(S)$ can be written as a finite disjoint union of sets in S.

We now wish to assign a 'correct' setting to talk about infinite coin tosses (or simple random walks on $\mathbb{Z}$). Let $\Omega = \{0,1\}^{\mathbb{N}}$, and we define

$$S = \{\{\mathbf{x} \mid x_1 = a_1,\, x_2 = a_2, \ldots, x_n = a_n\} \mid n \in \mathbb{N},\, a_i \in \{0,1\}\} \cup \{\emptyset\} \tag{1.22}$$

and let $\mathcal{F} = \sigma(S)$. Verify that S is a π-system; we can set

$$\mu(\{\mathbf{x} \mid x_1 = a_1,\, x_2 = a_2, \ldots, x_n = a_n\}) = \prod_{i=1}^{n} p^{a_i}(1-p)^{1-a_i}. \tag{1.23}$$

For now, assume that μ is a pre-measure on S. Then there'll exist a unique probability measure on $(\Omega, \sigma(S))$ that extends μ.

1.3 Measurable Maps and Random Variables

Note that a space $(\Omega, \mathcal{F})$ is called a *measurable space*.

Definition 1.16. A map $T : (\Omega, \mathcal{F}) \to (\tilde{\Omega}, \tilde{\mathcal{F}})$ between two measurable spaces is termed a *measurable function* if $T^{-1}(A) \in \mathcal{F}$ for all $A \in \tilde{\mathcal{F}}$. When $(\tilde{\Omega}, \tilde{\mathcal{F}}) = (\mathbb{R}, \mathcal{B})$, we call T a *random variable*.

In most situations, X is usually a topological space and $\mathcal{B}(X)$ is the Borel σ-algebra on X. For the case of $X = \mathbb{R}$, a random variable can be interpreted as a meaningful quantity which can be studied.

Proposition 1.17. *$T : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B})$ is measurable if and only if $T^{-1}(U) \in \mathcal{F}$ whenever U is open in $\mathbb{R}$.*

Proof. One direction is clear; for the converse direction, let $\tilde{\mathcal{F}} = \{W \subseteq \mathbb{R} \mid T^{-1}(W) \in \mathcal{F}\}$. It is relatively easy to show that $\tilde{\mathcal{F}}$ is a σ-algebra on $\mathbb{R}$. Thus, since $\mathcal{B} \subseteq \tilde{\mathcal{F}}$, we have $T^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{B}$. ■

Note that the image space being in $\mathbb{R}$ did not play a role in the above proof; for any image space $\tilde{\Omega}$, we can define $\tilde{\mathcal{F}} = \{W \subseteq \tilde{\Omega} \mid T^{-1}(W) \in \mathcal{F}\}$, and show that $\tilde{\mathcal{F}}$ is a σ-algebra on $\tilde{\Omega}$. Such a $\tilde{\mathcal{F}}$ is called the *pushforward σ-algebra* of $\mathcal{F}$ under T, and is the largest σ-algebra on $\tilde{\Omega}$ such that T is measurable. One can think in the converse way, where we are given a map $T : \Omega \to (\tilde{\Omega}, \tilde{\mathcal{F}})$; to make such a T measurable, one can simply equip Ω with the powerset σ-algebra 2^{Ω}, and then T is automatically measurable. Alternatively, we can define $\mathcal{F} = \{T^{-1}(A) \mid A \in \tilde{\mathcal{F}}\}$, and then T is measurable. This is called the *pullback σ-algebra* of $\tilde{\mathcal{F}}$ under T, and is the smallest σ-algebra on Ω such that T is measurable.

The most non-trivial yet basic example of a random variable is the *indicator function* of a set $A \in \mathcal{F}$, which is defined as $\mathbf{1}_A(\omega) = 1$ if $\omega \in A$, and 0 otherwise. It is easy to check that $\mathbf{1}_A$ is a random variable.

Lemma 1.18. *If $X, Y : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B})$ are random variables, then $X + Y$ is also a random variable. More generally, if $f : \mathbb{R}^2 \to \mathbb{R}$ is continuous, then $f(X, Y)$ is a random variable.*

August 7th.

Example 1.19. Consider the infinite simple random walk from before; for $a \in \mathbb{Z}$, define $\tau_a = \inf\{n > 0 \mid S_n = a\}$, the first hitting time of a. Then we claim that τ_a is measurable and, hence, a random variable. To see this, note that $\tau_a^{-1}((n, \infty)) = \{\omega \in \Omega \mid \tau_a(\omega) > n\} = \{\omega \in \Omega \mid S_1(\omega) \neq a, S_2(\omega) \neq a, \ldots, S_n(\omega) \neq a\} = \bigcap_{k=1}^{n} \{\omega \in \Omega \mid S_k(\omega) \neq a\}$, which is in $\mathcal{F}$ since S_k is measurable for all k. Thus, τ_a is measurable.

Definition 1.20. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $T : (\Omega, \mathcal{F}) \to (\tilde{\Omega}, \tilde{\mathcal{F}})$ be a measurable map. Then, the *pushforward measure* of $\mathbb{P}$ under T is the measure $\tilde{\mathbb{P}}$ on $(\tilde{\Omega}, \tilde{\mathcal{F}})$ defined by $\tilde{\mathbb{P}}(A) = \mathbb{P}(T^{-1}(A))$ for all $A \in \tilde{\mathcal{F}}$.

Indeed, one can show that the pushforward measure $\tilde{\mathbb{P}}$ is a probability measure on $(\tilde{\Omega}, \tilde{\mathcal{F}})$.

Definition 1.21. Given a Borel probability measure μ on $(\mathbb{R}, \mathcal{B})$, we define the *cumulative distribution function* of μ to be the function $F_\mu : \mathbb{R} \to [0, 1]$ defined by $F_\mu(x) = \mu((-\infty, x])$ for all $x \in \mathbb{R}$.

Proposition 1.22. *With μ as above, the following hold true.*

1. *F_μ is non-decreasing.*
2. *$\lim_{x\to\infty} F_\mu(x) = 1$ and $\lim_{x\to-\infty} F_\mu(x) = 0$.*
3. *F_μ is right-continuous.*

Proof.

1. If $x_1 \leq x_2$, then $(-\infty, x_1] \subseteq (-\infty, x_2]$, and hence $F_\mu(x_1) = \mu((-\infty, x_1]) \leq \mu((-\infty, x_2]) = F_\mu(x_2)$.
2. Note that $\mathbb{R} = \bigcup_{n=1}^\infty (-\infty, n]$; defining $A_k = (-\infty, k]$, we have $A_1 \subseteq A_2 \subseteq \cdots$ and $A_k \uparrow \mathbb{R}$. Since μ is a measure, $\mu(A_k) \uparrow \mu(\mathbb{R}) = 1$, and hence $\lim_{x\to\infty} F_\mu(x) = 1$. Similarly, $\lim_{x\to-\infty} F_\mu(x) = 0$.
3. Let x_n be a decreasing sequence of numbers such that $x_n \downarrow x$. Then, $(-\infty, x_1] \supseteq (-\infty, x_2] \supseteq \cdots$ and $(-\infty, x_n] \downarrow (-\infty, x]$. Since μ is a measure, $\mu((-\infty, x_n]) \downarrow \mu((-\infty, x])$, and hence $F_\mu(x_n) \downarrow F_\mu(x)$. Thus, F_μ is right-continuous.

■

Theorem 1.23. *Suppose $F : \mathbb{R} \to [0, 1]$ is a non-decreasing, right-continuous function such that $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$. Then, there exists a unique Borel probability measure μ on $(\mathbb{R}, \mathcal{B})$ such that $F = F_\mu$.*

Thus, there is a one-to-one correspondence between Borel probability measures on $(\mathbb{R}, \mathcal{B})$ and cumulative distribution functions. In general, it easier to study the cumulative distribution function of a probability measure than the measure itself.

1.3.1 The Pushfoward Measure

August 11th.

In many situations, we can model a random experiment in several ways. Let $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}$, and λ the Lebesgue measure. Define the random variables $X_1 = \mathbf{1}_{[0,1/2]}$ and $X_2 = \mathbf{1}_{[0,1/4]\cup[1/2,3/4]}$. Consider $\Gamma = (X_1, X_2) : \Omega \to \mathbb{R}^2$, and let $\mu = \lambda \circ \Gamma^{-1}$ be the pushforward measure of λ under Γ. The support of μ is the set $\Gamma(\Omega) = \{(1, 0), (0, 1), (1, 1), (0, 0)\}$. The measure at each of these points is given by $\mu(\{(1, 0)\}) = \lambda(\Gamma^{-1}(\{(1, 0)\})) = \lambda([0, 1/4]) = 1/4$, $\mu(\{(0, 1)\}) = \lambda(\Gamma^{-1}(\{(0, 1)\})) = \lambda([1/2, 3/4]) = 1/4$, $\mu(\{(1, 1)\}) = \lambda(\Gamma^{-1}(\{(1, 1)\})) = \lambda([1/4, 1/2]) = 1/4$, and $\mu(\{(0, 0)\}) = \lambda(\Gamma^{-1}(\{(0, 0)\})) = \lambda([3/4, 1]) = 1/4$. Thus, we have a probability measure on the support of μ. That is, we have modeled tossing a fair coin twice, without explicitly defining the sample space as $\{0, 1\}^2$. The probability space does not matter in most contexts; the random variables and their distributions are what matter.

Naturally, we can ask further questions. Whether the following models can exist: tossing a coin n times, and tossing a coin countably many times. The first question is easy to answer; we can define the k-th random variable X_k exactly as

$$X_k = \mathbf{1}_{[0,\frac{1}{2^k}]\cup[\frac{2}{2^k},\frac{3}{2^k}]\cup\cdots\cup[\frac{2^k-2}{2^k},\frac{2^k-1}{2^k}]}. \tag{1.24}$$

We leave it as an exercise to verify that $(X_1, X_2, \ldots, X_n)$ has the same distribution as tossing a fair coin n times.

Theorem 1.24. *Given any probability measure μ on $(\mathbb{R}, \mathcal{B})$, there exists a measurable map $T : ([0, 1], \mathcal{B}([0, 1]), \lambda) \to (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ such that $\mu = \lambda \circ T^{-1}$. That is, the pushforward measure of λ under T is μ.*

Before we proceed with this theorem, we show the following proposition.

Proposition 1.25. *For any Borel probability measures μ and ν on $(\mathbb{R}, \mathcal{B})$, if $F_\mu = F_\nu$, then $\mu = \nu$. That is, the cumulative distribution function uniquely determines the probability measure.*

Proof. Let μ and ν be the required measures such that $F_\mu = F_\nu$, or $\mu((-\infty, x]) = \nu((-\infty, x])$ for all $x \in \mathbb{R}$. Let $S = \{(-\infty, x] \mid x \in \mathbb{R}\}$, which is a π-system. Note that $\sigma(S) = \mathcal{B}$; thus, by the Sierpiński-Dynkin π-λ theorem, we have $\mu(A) = \nu(A)$ for all $A \in \mathcal{B}$. Hence, $\mu = \nu$. ■

We now prove **Theorem 1.23** and **Theorem 1.24** together.

Proof of theorems. Suppose $F : \mathbb{R} \to [0, 1]$ be a non-decreasing, right-continuous function such that $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$. We will show that there exists a measurable map $T : ([0, 1], \mathcal{B}([0, 1]), \lambda) \to (\mathbb{R}, \mathcal{B})$ such that $F_{\lambda\circ T^{-1}} = F$; note that this implies the second theorem by the previous proposition.

For such an F, we can define a generalized inverse T defined as $T(x) = \inf\{p \in \mathbb{R} \mid F(p) \geq x\}$. For any $x \in [0, 1]$, the set $\{p \in \mathbb{R} \mid F(p) \geq x\}$ is non-empty (since $\lim_{p\to\infty} F(p) = 1$) and has infimum, so T is well-defined. Moreover, T is non-decreasing and takes values in $\mathbb{R}$. To show measurability of T, note that for any $a \in \mathbb{R}$, we have

$$\{x \in [0, 1] \mid T(x) > a\} = \{x \in [0, 1] \mid F(a) < x\} = (F(a), 1] \cap [0, 1], \tag{1.25}$$

which is a Borel set in $[0, 1]$. Thus, T is measurable. We now verify that $F_{\lambda\circ T^{-1}} = F$; For any $x \in \mathbb{R}$, we have

$$\begin{aligned}(\lambda \circ T^{-1})((-\infty, x]) &= \lambda(T^{-1}((-\infty, x])) = \lambda(\{u \in [0, 1] \mid T(u) \leq x\}) \\ &= \lambda(\{u \in [0, 1] \mid \inf\{p \in \mathbb{R} \mid F(p) \geq u\} \leq x\}) = \lambda(\{u \in [0, 1] \mid F(x) \geq u\}) \\ &= \lambda([0, F(x)]) = F(x). \end{aligned}\tag{1.26}$$

Thus, $F_{\lambda\circ T^{-1}} = F$, establishing the existence of the desired measure. ■

1.4 Probability Measures on $\mathbb{R}$

August 14th.

We first wish to generalize the notion of the integral of a function with respect to the Lebesgue measure. A function $\varphi : [a, b] \to \mathbb{R}$ is said to be a *simple function* if it is of the form $\varphi = \sum_{i=1}^n a_i \mathbf{1}_{A_i}$ for some $n \in \mathbb{N}$, $a_i \in \mathbb{R}$, and $A_i \in \mathcal{B}$. That is, φ takes finitely many values, and the pre-image of each value is a Borel set. For a simple function φ, we define the integral of φ to be

$$\int_{[a,b]} \varphi(x)\, dx = \sum_{i=1}^n a_i \lambda(A_i). \tag{1.27}$$

In general, a function $g : [a, b] \to \mathbb{R}$ is said to be *Borel measurable* if $g^{-1}(A) \in \mathcal{B}$ for all Borel sets $A \subseteq \mathbb{R}$. Note that any simple function is Borel measurable. The integral of a non-negative Borel measurable function $g : [a, b] \to \mathbb{R}$ is then defined as

$$\int_{[a,b]} g(x)\, dx = \sup \left\{ \int_{[a,b]} \varphi(x)\, dx \;\middle|\; 0 \leq \varphi \leq g,\ \varphi \text{ is simple} \right\}. \tag{1.28}$$

One can infer that $\int(f + \alpha g) = \int f + \alpha \int g$, $g \leq f \implies \int g \leq \int f$, and $\int_{I_1\sqcup I_2} g = \int_{I_1} g + \int_{I_2} g$ for disjoint intervals I_1 and I_2, and non-negative Borel measurable functions f and g. We can extend the definition of the integral to all Borel measurable functions g by defining $\int g = \int g^+ - \int g^-$, where $g^+ = \max\{g, 0\}$ and $g^- = \max\{-g, 0\}$. Thus, we are fit to define the following.

Definition 1.26. A probability measure $\mu \in \mathcal{P}(\mathbb{R})$ is said to have a *density* f if $f : \mathbb{R} \to \mathbb{R}$ is a non-negative Borel measurable function such that $\int_{\mathbb{R}} f(x)\, dx = 1$ and $F_\mu(x) = \int_{-\infty}^x f(t)\, dt$ for all $x \in \mathbb{R}$.

Suppose $\mu \in \mathcal{P}(\mathbb{R})$ has a density f. Then, for any Borel set $A \subseteq \mathbb{R}$, we have

$$\mu(A) = \int_A f(x)\, dx. \tag{1.29}$$

Note that not every probability measure has a density; for example, the measure $\mu = \delta_0$, indicator of the set $\{0\}$, does not have a density. In fact, if there exists $p \in \mathbb{R}$ with $\mu(\{p\}) > 0$, then μ does not have a density. This is because if μ has a density f, then $\mu(\{p\}) = \int_{\{p\}} f(x)\,dx = 0$, which is a contradiction. Thus, any probability measure with a density must be non-atomic; that is, it cannot assign positive measure to any singleton set. However, the converse is not true; there exist non-atomic probability measures that do not have a density. For example, consider the Cantor measure μ on $[0,1]$, which is non-atomic, but does not have a density.

We now wish to define a metric on the space of probability measures on $\mathbb{R}$.

1. For any $\mu, \nu \in \mathcal{P}(\mathbb{R})$, we define the *total variation distance* between μ and ν to be

$$D_1(\mu,\nu) = d_{TV}(\mu,\nu) := \sup_{A\in\mathcal{B}} |\mu(A) - \nu(A)|\,. \tag{1.30}$$

2. The CDFs may be utilized to define

$$D_2(\mu,\nu) := \sup_{x\in\mathbb{R}} |F_\mu(x) - F_\nu(x)|\,. \tag{1.31}$$

Both of these metrics are fine as they are, but fail in being 'nice' with respect to convergence of measures; the sequence of measures $\{\mu_n = \delta_{1/n}\}_{n=1}^\infty$ fails to converge to δ_0 in either of these metrics, even though we expect it to converge intuitively.

August 18th.

Hereforth, we define the *Levy distance* on $\mathcal{P}(\mathbb{R})$ as

$$d(\mu,\nu) := \inf\{\epsilon > 0 \mid F_\mu(t+\epsilon) + \epsilon \geq F_\nu(t),\ F_\nu(t+\epsilon) + \epsilon \geq F_\mu(t) \text{ for all } t \in \mathbb{R}\}. \tag{1.32}$$

In fact, this can be generalized to a metric on Borel probability measures on any metric space $(X, \mathcal{D})$ as follows (this is termed the *Levy-Prokhorov metric*).

$$d_{LP}(\mu,\nu) := \inf\{\epsilon > 0 \mid \mu(A+\epsilon) + \epsilon \geq \nu(A),\ \nu(A+\epsilon) + \epsilon \geq \mu(A) \text{ for all closed } A \subseteq X\} \tag{1.33}$$

where the *ϵ-fattening* of A is defined as

$$A + \epsilon := \{x \in X \mid \text{ there exists } y \in A \text{ such that } \mathcal{D}(x,y) < \epsilon\}. \tag{1.34}$$

Note that these two metrics are not the same on $\mathcal{P}(\mathbb{R})$, but are equivalent. Also, via the Levy distance, we can show $d(\delta_{1/n}, \delta_0) = 1/n$, showing that $\delta_{1/n} \to \delta_0$ in this metric.

August 25th.

Theorem 1.27. *Suppose $\mu, \mu_1, \mu_2, \ldots \in \mathcal{P}(\mathbb{R})$ are Borel probability measures. Then $d(\mu_n, \mu) \xrightarrow{n\to\infty} 0$ if and only if $F_{\mu_n}(x) \xrightarrow{n\to\infty} F_\mu(x)$ for all continuity points $x \in \mathbb{R}$ of F_μ.*

Proof. We first show the forward implication. Suppose $d(\mu_n, \mu) \to 0$. Then, for any $\epsilon > 0$, there exists $N_\epsilon \in \mathbb{N}$ such that $d(\mu_n, \mu) < \epsilon$ for all $n \geq N_\epsilon$. Now fix $\epsilon > 0$ and $x \in \mathbb{R}$. Then, for all $n \geq N_\epsilon$, we have $F_\mu(x+\epsilon) + \epsilon \geq F_{\mu_n}(x)$. We can replace the right hand side with the limit superior to get

$$F_\mu(x+\epsilon) + \epsilon \geq \limsup_{n\to\infty} F_{\mu_n}(x) \qquad \text{for all } \epsilon > 0. \tag{1.35}$$

This implies that $F_\mu(x) \geq \limsup_{n\to\infty} F_{\mu_n}(x)$. Similarly, for all $n \geq N_\epsilon$, we have $F_{\mu_n}(x) + \epsilon \geq F_\mu(x-\epsilon)$. By a similar argument, we have $\liminf_{n\to\infty} F_{\mu_n}(x) \geq F_\mu(x)$ if x is a continuity point of F_μ. Thus, we have shown that $\limsup_{n\to\infty} F_{\mu_n}(x) \leq F_\mu(x) \leq \liminf_{n\to\infty} F_{\mu_n}(x)$ or $F_{\mu_n}(x) \to F_\mu(x)$ for all continuity points.

For the converse, if $\mathcal{C}$ denotes the set of continuity points of F_μ, then $\mathcal{C}$ is cocountable and hence dense in $\mathbb{R}$. Now fix $\varepsilon \in (0,1)$. Let $A, B \in \mathbb{R}$ be numbers such that $F_\mu(A) < \varepsilon$ and $F_\mu(B) \geq 1-\varepsilon$. Pick points $\{x_i\}_{i=0}^m$ in $[A,B] \cap \mathcal{C}$ such that $A = x_0 < x_1 < \cdots < x_m = B$ and $x_{i+1} - x_i < \varepsilon$ for all i. By the choice of the x_i's, we have $F_{\mu_n}(x_i) \to F_\mu(x_i)$ for all i, and so there exists $N_\varepsilon \in \mathbb{N}$ such that $|F_{\mu_n}(x_i) - F_\mu(x_i)| < \varepsilon$ for all $n \geq N_\varepsilon$ and all i. Now fix $t \in [A,B]$; so $t \in [x_i, x_{i+1}]$ for some i. Then, for all $n \geq N_\varepsilon$, we have

$$F_{\mu_n}(t+\varepsilon) \geq F_{\mu_n}(x_{i+1}) \geq F_\mu(x_{i+1}) - \varepsilon \geq F_\mu(t) - \varepsilon \implies F_{\mu_n}(t+\varepsilon) + \varepsilon \geq F_\mu(t) \tag{1.36}$$

for all $n \geq N_\varepsilon$ and $t \in [A, B]$. If $t > B$, then both $F_{\mu_n}(t)$ and $F_\mu(t)$ are at least $1 - 2\varepsilon$, and hence $F_{\mu_n}(t+\varepsilon)+\varepsilon \geq F_\mu(t)$. If $t < A$, then both $F_{\mu_n}(t)$ and $F_\mu(t)$ are at most 2ε, and hence $F_{\mu_n}(t+\varepsilon)+\varepsilon \geq F_\mu(t)$. Thus, we have shown that for all $\varepsilon > 0$, there exists $N_\varepsilon \in \mathbb{N}$ such that for all $n \geq N_\varepsilon$, we have

$$F_{\mu_n}(t + \varepsilon) + \varepsilon \geq F_\mu(t) \quad \text{for all } t \in \mathbb{R} \implies d(\mu_n, \mu) \leq 2\varepsilon. \tag{1.37}$$

■

1.5 Expectation

Expectation, in our new setting, corresponds to integration in a probability space. To this end, we start with simple functions.

Definition 1.28. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A measurable function $S : \Omega \to \mathbb{R}$ is called a *simple random variable* if it takes finitely many values; that is, $S(\Omega) = \{s_1, \ldots, s_n\}$ for some $n \in \mathbb{N}$.

In this case, we can write $S = \sum_{i=1}^n c_i \mathbf{1}_{A_i}$ for some $c_i \in \mathbb{R}$ and $A_i \in \mathcal{F}$. This representation need not be unique; however, note that any such simple function can be written in a unique way as

$$S = \sum_{i=1}^{n} \beta_i \mathbf{1}_{B_i} \tag{1.38}$$

where $\beta_i \in S(\Omega)$ are unequal and B_i's are measurable sets that are mutually disjoint.

Definition 1.29. The *expectation of a simple random variable* $S = \sum_{i=1}^n c_i \mathbf{1}_{A_i}$ is defined as

$$\mathbb{E}[S] := \sum_{i=1}^{n} c_i \mathbb{P}(A_i). \tag{1.39}$$

Since this representation is not unique, one can verify that the expectation is well-defined by using the unique representation of S as $\sum_{i=1}^n \beta_i \mathbf{1}_{B_i}$. Also, if S and T are simple random variables, then $S+T$ is also a simple random variable, and $\mathbb{E}[S + T] = \mathbb{E}[S] + \mathbb{E}[T]$. Similarly, for any $\alpha \in \mathbb{R}$, we have $\mathbb{E}[\alpha S] = \alpha \mathbb{E}[S]$. If $S \geq 0$, then $\mathbb{E}[S] \geq 0$. We can now extend the definition of expectation to all non-negative random variables, and then to all random variables.

Definition 1.30. The *expectation of a non-negative random variable* $X : \Omega \to [0, \infty)$ is defined as

$$\mathbb{E}[X] := \sup\{\mathbb{E}[S] \mid 0 \leq S \leq X, S \text{ is a simple random variable}\}. \tag{1.40}$$

If X is a general random variable, we can write $X = X^+ - X^-$, where $X^+ = \max\{X, 0\}$ and $X^- = \max\{-X, 0\}$. If $\mathbb{E}[X^+] < \infty$ or $\mathbb{E}[X^-] < \infty$, then we define the *expectation of (the) random variable* X as

$$\mathbb{E}[X] := \mathbb{E}[X^+] - \mathbb{E}[X^-]. \tag{1.41}$$

Proposition 1.31. *Let X and Y be non-negative random variables. Then, the following hold true.*

1. $\mathbb{E}[X] \geq 0$.
2. $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.

Proof. The first part is clear from the definition since $S \equiv 0$ is a simple random variable such that $0 \leq S \leq X$, so $\mathbb{E}[X] \geq \mathbb{E}[S] = 0$. For the second part, let S and T be simple random variables such that $0 \leq S \leq X$ and $0 \leq T \leq Y$. Then, $S + T$ is a simple random variable such that $0 \leq S + T \leq X + Y$, and hence $\mathbb{E}[X + Y] \geq \mathbb{E}[S + T] = \mathbb{E}[S] + \mathbb{E}[T]$. Taking supremum over all such S and T, we get $\mathbb{E}[X + Y] \geq \mathbb{E}[X] + \mathbb{E}[Y]$. For the reverse inequality, let U be a simple random variable such that $0 \leq U \leq X + Y$. Then, we can write $U = U_1 + U_2$ where $U_1 = U\mathbf{1}_{\{X>0\}}$ and $U_2 = U\mathbf{1}_{\{Y>0\}}$. Then,

$0 \le U_1 \le X$ and $0 \le U_2 \le Y$, and hence $\mathbb{E}[U] = \mathbb{E}[U_1] + \mathbb{E}[U_2] \le \mathbb{E}[X] + \mathbb{E}[Y]$. Taking supremum over all such U, we get $\mathbb{E}[X+Y] \le \mathbb{E}[X] + \mathbb{E}[Y]$. Thus, we have equality. ■

We also have the monotone converge theorem, of which we state a basic version first.

Theorem 1.32. *Suppose $X \ge 0$ is a random variable, and $\{\varphi_n\}_{n=1}^{\infty}$ is a sequence of simple random variables such that $0 \le \varphi_n \uparrow X$. Then, $\mathbb{E}[\varphi_n] \uparrow \mathbb{E}[X]$.*

Theorem 1.33 (The *monotone convergence theorem*)**.** *Suppose $X \ge 0$ is a random variable, and $\{X_n\}_{n=1}^{\infty}$ is a sequence of non-negative random variables such that $0 \le X_n \uparrow X$. Then, $\mathbb{E}[X_n] \uparrow \mathbb{E}[X]$.*

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